PERSONAL INFORMATION

Family name	Gandhi
First name	Yogesh
Date of Birth	16/12/1991
ORCID code	0000-0002-8330-8515

EDUCATION

2020–2024 **PhD in Aerospace Engineering**, *UNIBO*, Forlì, Italy
Thesis: Geometry projection method for additively manufacture variable stiffness continuous fiber-reinforced polymer structures—A unified topology optimization approach for multi-layered composite laminates.

2015–2017 **Masters in Aerospace Engineering**, *UNIBO*, Forlì, Italy
Thesis: Motion planning and control for Differential Drive Wheel Mobile Robot.

2010–2014 **Bachelors in Mechanical Engineering**, *PESIT*, Bengaluru, India
Thesis: Piezoelectric Energy Harvesting in Micro-Air Vehicle (MAV).

PREVIOUS EXPERIENCE

01/24–06/24 **POST-DOCTORAL RESEARCHER**, *University of Bologna (UNIBO)*, Forlì, Italy

1. **Topology optimization (TO)** methods for continuous fiber-reinforced polymer (CFRP) components within the **ASCENT project**, designed for 3D printers (3DP) such as: Markforged and 9tlabs.
2. Design, analysis, and 3D printing of the Emilia V solar-powered vehicle, contributing to its victory in the Sasol 2024 cruiser challenge.
3. Stress-based design framework to 3D-print Ferrari 296GTB wing mechanism parts within the **OptiFiber project** in collaboration with **FERRARI**.

04/18–02/20 **RESEARCH ASSOCIATE**, *University of Parma*, Parma, Italy

1. Modeled the thermo-mechanical response of Shape Memory Alloy (SMA) using the UMAT subroutine and developed a fiber optimization method to predict bistable laminate designs.
2. Numerical modeling for predicting the initiation and evolution of **delamination in composites using the VCCT and cohesive elements technique**.

RESEARCH SECONDMENTS

06/23–10/24 **Research stay & Collaboration**, *Politecnico di Torino*, Torino, Italy
Developed a new TO method leveraging the Carrera Unified Formulation (CUF) to design **Multilayered composite laminates**.

05/23–06/25 **Research stay & Collaboration**, *Università degli Studi di Padova*, Vicenza, Italy
Analyze and optimize process parameters of 9tlabs 3DP to improve structural integrity through post-processing steps.

Collaborated with Prof. Paolo Carraro on the **ASCENT and OptiFiber projects** to design a framework for 9tlabs 3DP by leveraging experimental data, conducting material characterization, and validating models.

08/21–12/24 **Research Collaborator**, *Delft University of Technology*, Delft, The Netherlands
Developed a new CAD-based TO method to optimize the layout and orientation of continuous fibers in composite laminates.

Collaborated with scholars in computational solid mechanics and structure optimization: Prof. Julián Norato and Prof. Alejandro Argón.

SHORT-TERM RESEARCH EXPERIENCE

04/17–09/17 **Visiting Graduate Researcher**, *Coventry University*, Coventry, UK
Developed algorithms for path planning (A-star and D-star) and obstacle avoidance for the rover that adhered to kinematic and dynamic limitations.

Trajectory traversal with an online non-linear controller based on a genetic algorithm (Particle Swarm) Optimization.

08/14–05/15 **Research Assistant**, *CORI, PESIT*, Bengaluru, India
The FSI model analyzes MAV wing flutter to validate my bachelor's thesis formulation.

SCIENTIFIC CAREER DESCRIPTION

PhD & Postdoctoral Excellence: I completed my PhD in Aerospace Engineering (2024), where I proposed TO methods for 3D-printed continuous fiber-reinforced polymers. This work led to secure funding for ASCENT and OptiFiber projects. **Interdisciplinary Integration & Independence:** My PhD journey was marked by a series of research secondments under the guidance of Prof. Julián Norato (USA) and Prof. Alfonso Pagani (Italy). This required rapid mastery of advanced mathematical concepts (e.g., Structure Optimization, Carrera Unified Formulation), confirming my capacity to thrive in new fields. Importantly, peer-reviewed papers spanning smart materials (Materials, 2019), composites, additive manufacturing, and structure optimization—evidence of autonomous scientific agility across engineering, physics, and materials science. **Translational Research Leadership:** In my first postdoc at UNIBO (2024), I successfully transitioned to applied industrial R&D. Here, I co-led projects that delivered 3D-printed components for the Emilia V solar vehicle, which emerged as the winner of the Sasol 2024 Cruiser Challenge, and for Ferrari 296GTB. This leadership role underscored my ability to deliver successful projects in an industrial setting. **Foundation for Teno-InGrowth:** My expertise bridges computational mechanics (FEM, phase-field-adjacent TO), 3D printing process optimization (Markforged, 9tlabs), and multidisciplinary collaboration—directly aligning with Teno-InGrowth's mechanobiology-driven scaffold design, positioning me to: Integrate dynamic tissue growth models with TO (OBJ1-2), Translate designs into FDA-compliant implants via CETIM/BioTech Dental (OBJ3), and Establish a spin-off (inSilicoTendon) for scalable medical device innovation.

Journal articles

Y. Gandhi, A. Aragón, J. Norato, and G. Minak. A geometry projection method for the topology optimization of additively manufactured variable-stiffness composite laminates. *Computer Methods in Applied Mechanics and Engineering*, 2025. doi: https://doi.org/10.1016/j.cma.2024.117663. URL `https://www.sciencedirect.com/science/article/pii/S0045782524009174`.

Y. Gandhi and G. Minak. A review on topology optimization strategies for additively manufactured continuous fiber-reinforced composite structures. *Applied Sciences*, 2022. doi: 10.3390/app122111211. URL `http://dx.doi.org/10.3390/app122111211`.

Y. Gandhi, A. Pirondi, and L. Collini. Optimal design of shape memory alloy composite under deflection constraint. *Materials*, 2019. doi: 10.3390/ma12111733. URL `https://doi.org/10.3390%2Fma12111733`.

Conference proceedings

Y. Gandhi, J. Norato, A. Pavlovic, and G. Minak. A geometry projection method for designing and optimizing additively manufactured variable-stiffness composite laminates. *Acta Polytechnica CTU Proceedings*, 48, 2024. doi: 10.14311/app.2024.48.0022. URL `http://dx.doi.org/10.14311/app.2024.48.0022`.

A. Pirondi, Y. Gandhi, and L. Collini. Strategies for modelling and optimization of bi-stable composite laminates actuated by embedded shape memory alloy wires. *Procedia Structural Integrity*, 24, 2019. doi: 10.1016/j.prostr.2020.02.042. URL `https://doi.org/10.1016%2Fj.prostr.2020.02.042`.

Y. Gandhi, A. Pirondi, and L. Collini. Analysis of bistable composite laminate with embedded SMA actuators. *Procedia Structural Integrity*, 12, 2018. doi: 10.1016/j.prostr.2018.11.075. URL `https://doi.org/10.1016%2Fj.prostr.2018.11.075`.

CORE COMPETENCIES

LANGUAGES	ENGLISH (C1)	ITALIAN (B2)
COMPUTATIONAL TOOLS	Programming	MATLAB ‖ PYTHON ‖ FORTRAN
	CAE	ANSYS ‖ ABAQUS ‖ SOLIDWORKS
	Others	LINUX ‖ Git ‖ LaTeX
Composites	FEM/FEA » Linear Elastic | NonLinear | | Thermal Stress | Fluid Dynamics | Material Behaviour | Structural Optimization	
	Multiscale Modelling | Failure & Damage Mechanisms	
	Mechanical Testing | Material Characterization | 3D Printing	

MENTORING EXPERIENCE

07/2024–Present **Doctoral candidate** **Co-supervisor**	Giulia Babbi (2^{nd} year), Aerospace Engineering, University of Bologna, Italy Develop a design framework for Markforged 3D-printed parts utilizing experimental data from Emilia V's components and performing material characterization and experiments validation.
05/2024–12/2024 **Master student** **Co-supervisor**	**Alberto Ancarani**, Mechanical Engineering, University of Bologna, Italy Development of a two-dimensional **stress-based topology optimization method** for the design of additively manufacturable structures.

CONFERENCE PRESENTATIONS

12-14 Sept. 2024 **Mathematical Institute of SANU** **Belgrade, SE**	2^{nd} International Conference of Mathematical Modelling in Mechanics and Eng. A Geometry Projection Method with Length Constraint for Designing Monolithic Structures made of CFRP.
03-06 Sept. 2024 **University of Bologna** **Ravenna, IT**	27^{th} International Conference on Composite Structures Experimental and Numerical Investigation of Additively Manufactured CFRP structures.
05-07 June 2024 **Universita di Cusano** **Rome, IT**	19^{th} Youth Symposium on Experimental Solid Mechanics A General Design and Optimization framework for Additively Manufacturable Variable-Stiffness Composite Laminates.
06-09 Sept. 2023 **University of Genova** **Italy**	52^{nd} Convegno Nazionale AIAS Geometry Projection Method for Designing and Manufacturing Multi-Layered Composite Laminate.
11-14 June 2023 **Czech Technical University** **Prague, CZ**	18^{th} Youth Symposium on Experimental Solid Mechanics Geometry Projection Method for Designing and Manufacturing Variable-Stiffness Composite Laminate.
19-22 July 2022 **University of Porto** **Porto, PT**	25^{th} International Conference on Composite Structures Hybrid Geometry Projection Method for Fiber-Reinforced Composite Laminate using Lamination Parameters.

OTHER POSITIONS & CERTIFICATIONS

TEACHING EXPERIENCE	10/2019 (15 hrs.) MSc.–Automotive Eng. Dallara Academy	Numerical modeling of fiber-reinforced composite **failure** and **delamination** in ABAQUS.
	10/2023 & 10/2024 (12 hrs.) MSc.–Mechanical Eng. University of Bologna	Theory of TO algorithms and their implementation using MATLAB
JOURNALS' REVIEWER	Composite Structures · Additive Manufacturing	
SCIENTIFIC TRAINING	4th Int. School On Fatigue and Damage Mechanics of Composite Materials. Multiscale Techniques for NonLinear Analysis of Composite Structures. Int. Symposium on Multi-Scale Mechanics of Composites.	